

ASSIGNMENT NO.3

Create a menu driven program for Encryption and Decryption of Text files, user selects Encryption the text file should be read and converted to a non readable format and stored in a file. When he selects Decrypt, the encrypted file should be decrypted to its original format.

Code:

Main.java:

```
public class Program {

    public static void main(String[] args) {
        int choice;
        do {
            System.out.println("\n=====");
            System.out.println(" FILE ENCRYPTION / DECRYPTION");
            System.out.println("=====");
            System.out.println("1. Encrypt File");
            System.out.println("2. Decrypt File");
            System.out.println("3. Exit");
            System.out.println("=====");

            System.out.print("Enter your choice: ");
            choice = ConsoleInput.getInt();

            switch (choice) {
                case 1:
                    Encrypt.encryptFile();
                    break;
                case 2:
                    Decrypt.decryptFile();
                    break;
                case 3:
                    System.out.println("Exit");
                    break;
                default:
                    System.out.println("Invalid choice");
            }
        } while (choice != 3);
    }
}
```

Encrypt.java:

```
import java.io.File;
import java.io.FileInputStream;
import java.io.FileNotFoundException;
import java.io.FileOutputStream;
import java.io.IOException;

public class Encrypt {
    static void encryptFile() {
        FileInputStream inputStream = null;
        FileOutputStream outputStream = null;
        try {
            File inputFile = new File("D:\\JavaSE_17\\data.txt");
            File encryptedFile = new File("D:\\JavaSE_17\\encrypted.txt");

            inputStream = new FileInputStream(inputFile);
            outputStream = new FileOutputStream(encryptedFile);

            byte[] data = inputStream.readAllBytes();
            byte key = 5;

            for (int i = 0; i < data.length; i++) {
                data[i] = (byte) (data[i] ^ key);
            }
            outputStream.write(data);
            System.out.println("File Encrypted Successfully");
            System.out.println("Encrypted file: D:\\JavaSE_17\\encrypted.txt");
        } catch (FileNotFoundException e) {
            System.out.println("File not found");
        } catch (IOException e) {
            System.out.println("Error while encrypting file");
        } finally {
            try {
                if (inputStream != null)
                    inputStream.close();

                if (outputStream != null)
                    outputStream.close();
            } catch (IOException e) {
                e.printStackTrace();
            }
        }
    }
}
```

```
public class Decrypt {
    static void decryptFile() {
        FileInputStream inputStream = null;
        FileOutputStream outputStream = null;
        try {
            File encryptedFile = new File("D:\\JavaSE_17\\encrypted.txt");
            File decryptedFile = new File("D:\\JavaSE_17\\decrypted.txt");

            inputStream = new FileInputStream(encryptedFile);
            outputStream = new FileOutputStream(decryptedFile);

            byte[] data = inputStream.readAllBytes();
            byte key = 5;

            for (int i = 0; i < data.length; i++) {
                data[i] = (byte) (data[i] ^ key);
            }

            outputStream.write(data);
            System.out.println("File Decrypted Successfully");
            System.out.println("Decrypted file: D:\\JavaSE_17\\decrypted.txt");

        } catch (FileNotFoundException e) {
            System.out.println("Encrypted file not found");
        } catch (IOException e) {
            System.out.println("Error while decrypting file");
        } finally {
            try {
                if (inputStream != null)
                    inputStream.close();

                if (outputStream != null)
                    outputStream.close();
            }
        }
    }
}
```

```
        } catch (IOException e) {  
            e.printStackTrace();  
        }  
    }  
}
```